

School of Computer Science and Engineering

Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lecture	Experiment	In Class / Take Home
Week : (23/02/26 to 1/03/26)	1	Minimum number of swaps You are given two binary strings s1 and s2 of equal length, and the task is to find the minimum number of swaps required to make them identical. The only allowed operation is swapping characters between the two strings (i.e., you can swap s1[i] with s2[j], but not within the same string). If it is impossible to make the two strings equal through such swaps, return -1. Examples: Input: s1 = "1100", s2 = "1111" Output: 1 Explanation: Swap the character at index 2 of s1 with the character at index 3 of s2. After this swap both strings become "1110", so only 1 swap is needed. Input: s1 = "00011", s2 = "11001" Output: -1 Explanation: It is impossible to make both strings equal, so the answer is -1. Link: https://www.geeksforgeeks.org/problems/minimum-number-of-swaps--164524/1?page=1&category=Strings&sortBy=latest	In Class
	1	Min Add to Make Parentheses Valid You are given a string s consisting only of the characters '(' and ')'. Your task is to determine the minimum number of parentheses (either '(' or ')') that must be inserted at any positions to make the string s a valid parentheses string. A parentheses string is considered valid if: - Every opening parenthesis '(' has a corresponding closing parenthesis ')' - Every closing parenthesis ')' has a corresponding opening parenthesis '(' - Parentheses are properly nested. Examples: Input: s = "(()(" Output: 2 Explanation: There are two unmatched '(' at the end, so we need to add two ')' to make the string valid. Input: s = ")))" Output: 3 Explanation: Three '(' need to be added at the start to make the string valid. Input: s = ")()()" Output: 1 Explanation: The very first ')' is unmatched, so we need to add one '(' at the beginning. Link: https://www.geeksforgeeks.org/problems/min-add-to-make-parentheses-valid/1?page=1&category=Strings&sortBy=latest	In Class
	1	Score of Parentheses String Given a string s consisting of balanced parentheses, calculate the score of the string based on the following rules:	Take Home

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		"()" has a score of 1. "AB" has a score of A + B, where A and B are balanced parentheses strings. "(A)" has a score of 2 × score(A), where A is a balanced parentheses string. Note: Test cases are generated such that the score will fit within a 32-bit integer. Examples: Input: s = "()()" Output: 2 Explanation: The string str is of the form "ab", that makes the total score = (score of a) + (score of b) = 1 + 1 = 2. Input: s = "(())" Output: 6 Explanation: The string str is of the form "(a(b))" which makes the total score = 2 × ((score of a) + 2 × (score of b)) = 2 × (1 + 2 × (1)) = 6. Input: s = "((()))" Output: 4 Explanation: The string str is of the form "((a))" which makes the total score = 2 × (2 × (score of a)) = 2 × (2 × (1)) = 4. Link: https://www.geeksforgeeks.org/problems/score-of-parentheses-string/1?page=1&category=Strings&sortBy=latest	
	1	Difference Check Given an array arr[] of time strings in 24-hour clock format "HH:MM:SS", return the minimum difference in seconds between any two time strings in the arr[]. The clock wraps around at midnight, so the time difference between "23:59:59" and "00:00:00" is 1 second. Examples: Input: arr[] = ["12:30:15", "12:30:45"] Output: 30 Explanation: The minimum time difference is 30 seconds. Input: arr[] = ["00:00:01", "23:59:59", "00:00:05"] Output: 2 Explanation: The time difference is minimum between "00:00:01" and "23:59:59". Link: https://www.geeksforgeeks.org/problems/difference-check/1?page=1&category=Strings&sortBy=latest	Take Home
	1	Sort by Frequency Given a string s, the task is to arrange the string according to the frequency of each character, in ascending order. If two elements have the same frequency, then they are sorted in lexicographical order. Examples: Input: s = "geeksforgeeks" Output: forggkkssee	Take Home

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		Explanation: All the characters with minimum frequency will occur first and the one with same frequency will be arranged lexicographically. Input: s = "abc" Output: abc Explanation: The frequency is one for all characters hence they'll be arranged lexicographically. Link: https://www.geeksforgeeks.org/problems/sort-string-according-to-increasing-frequency/1?page=1&category=Strings&sortBy=latest	
	1	Sort an array of strings according to string lengths You are given an array arr[] of strings. You have to sort the array in ascending order based on the lengths of the strings. If two strings have the same length, maintain their original relative order. Examples: Input: arr[] = ["GeeksforGeeks", "I", "from", "am"] Output: ["I", "am", "from", "GeeksforGeeks"] Explanation: The strings are sorted in increasing order of their lengths, starting from the shortest string "I" to the longest one "GeeksforGeeks". Input: arr[] = ["You", "are", "beautiful", "looking"] Output: ["You", "are", "looking", "beautiful"] Explanation: The strings are sorted by length: "You", "are", "looking", and then "beautiful", with the shortest words appearing first and the longest last. Link: https://www.geeksforgeeks.org/problems/sort-an-array-of-strings-according-to-string-lengths/1?page=1&category=Strings&sortBy=latest	Take Home
	1	Count pairs of strings with one mismatch Given an array arr[] of strings of equal length, where each string consists of lowercase English letters, find the total number of pairs of strings that differ in exactly one character position. Note: Two strings differ in exactly one position if they have the same length and differ in exactly one index that is, all characters except one are the same. Examples : Input: arr[] = ["abc", "abd", "bbd"] Output: 2 Explanation: The valid pairs are: 1. ("abc", "abd") → differ at index 2 2. ("abd", "bbd") → differ at index 0 Input: arr[] = ["def", "deg", "dmf", "xef", "dxg"] Output: 4 Explanation: The valid pairs are: 1. ("def", "deg") → differ at index 2 2. ("def", "dmf") → differ at index 1 3. ("def", "xef") → differ at index 0 4. ("deg", "dxg") → differ at index 1	Take Home

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		Input: arr[] = ["bcde", "bcd", "bdce"] Output: 0 Explanation: No pair differs at exactly one position. Link: https://www.geeksforgeeks.org/problems/count-pairs-of-strings-with-one-mismatch/1?page=1&category=Strings&sortBy=latest	
Week : (23/02/26 to 1/03/26)	2	Winner of an election Given a lowercase string array arr[]. Each element in the array represents a vote cast for a candidate. Return the name of the candidate who received the maximum number of votes and the count of votes he received. In case of a tie between two or more candidates, return the lexicographically smallest name. Note: Return an array of strings, the winning candidate name as the first element and the vote count as the second element (typecast the count to string). Examples : Input: arr[] = ["john", "johnny", "jackie", "johnny", "john", "jackie", "jamie", "jamie", "john", "johnny", "jamie", "johnny", "john"] Output: ["john", "4"] Explanation: "john" has 4 votes casted for him, but so does "johnny". "john" is lexicographically smaller, so we print "john" and the votes he received. Input: n = 3 arr[] = ["andy", "blake", "clark"] Output: ["Andy", "1"] Explanation: All the candidates get 1 votes each. We print "andy" as it is lexicographically smaller. Link: https://www.geeksforgeeks.org/problems/winner-of-an-election-where-votes-are-represented-as-candidate-names-1587115621--162858/1?page=2&category=Strings&sortBy=latest	In Class
	2	Check if Permutation is Substring Given two strings txt and pat having lowercase letters, the task is to check if any permutation of pat is a substring of txt. Examples: Input: txt = "geeks", pat = "eke" Output: true Explanation: "eek" is a permutation of "eke" which exists in "geeks". Input: txt = "programming", pat = "rain" Output: false Explanation: No permutation of "rain" exists as a substring in "programming". Link: https://www.geeksforgeeks.org/problems/check-if-permutation-is-substring/1?page=1&category=Strings&sortBy=latest	In Class

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	2	Lexicographically Largest String After K Deletions Given a string s consisting of lowercase English letters and an integer k, your task is to remove exactly k characters from the string. The resulting string must be the largest possible in lexicographical order, while maintain the relative order of the remaining characters. Examples: Input: s = "ritz", k = 2 Output: tz Explanation: By removing two characters in all possible ways, we get: "ri", "rt", "rz", "it", "iz", and "tz". Among these, "tz" is lexicographically largest string. Input: s = "zebra", k = 3 Output: zr Explanation: Removing "e", "b", and "a" results in "zr", which is lexicographically largest string. Link: https://www.geeksforgeeks.org/problems/lexicographically-largest-string-after-deleting-k-characters/1?page=1&category=Strings&sortBy=latest	Take Home
	2	Count Unique Vowel Strings You are given a lowercase string s, determine the total number of distinct strings that can be formed using the following rules: - Identify all unique vowels (a, e, i, o, u) present in the string. - For each distinct vowel, choose exactly one of its occurrences from s. If a vowel appears multiple times, each occurrence represents a unique selection choice. - Generate all possible permutations of the selected vowels. Each unique arrangement counts as a distinct string. Return the total number of such distinct strings. Examples: Input: s = "aeiou" Output: 120 Explanation: Each vowel appears once, so the number of different strings can form is $5! = 120$. Input: s = "ae" Output: 2 Explanation: Pick a and e, make all orders → "ae", "ea". Input: s = "aacidf" Output: 4 Explanation: Vowels in s are 'a' and 'i', Pick each 'a' once with a single 'i', make all orders → "ai", "ia", "ai", "ia". Link: https://www.geeksforgeeks.org/problems/count-unique-vowel-strings/1?page=1&category=Strings&sortBy=latest	Take Home

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	2	Count Even Letters You are given a string s consisting of lowercase english letters. Your task is to count how many distinct characters appear an even number of times in the string. Examples: Input: s = "abacaba" Output: 2 Explanation: The frequency of a is 4, b is 2 and c is 1 so there are 2 characters with even frequency. Input: s = "zzacccz" Output: 0 Explanation: The frequency of z is 3, a is 1 and c is 3 so there are no characters with even frequency. Link: https://www.geeksforgeeks.org/problems/count-even-letters/1?page=1&category=Strings&sortBy=latest	Take Home
	2	Shortest substring containing all vowels You are given two strings, s1 and s2, where s1 contains distinct lowercase vowels (a, e, i, o, u), and s2 contains lowercase English letters. Your task is to find the length of the shortest contiguous substring within s2 that contains all the vowels present in s1 at least once, in any order. Note: If there is no such substring in s2, return -1. Examples: Input: s1 = "ae", s2 = "acbaudeq" Output: 4 Explanation: The shortest substring of "acbaudeq" that contains both vowels 'a' and 'e' from s1 = "ae" is "aude", which has length 4. Input: s1 = "iou", s2 = "iuiouiu" Output: 3 Explanation: The shortest substring of "iuiouiu" that contains all vowels 'i', 'o', and 'u' from s1 = "iou" is "oiu", which has length 3. Input: s1 = "aeiou", s2 = "uoiee" Output: -1 Explanation: The string s2 = "uoiee" is missing the vowel 'a' from s1 = "aeiou", so no substring can contain all required vowels, and the answer is -1. Link: https://www.geeksforgeeks.org/problems/shortest-substring-containing-all-vowels/1?page=1&category=Strings&sortBy=latest	Take Home
	2	Balancing Consonants and Vowels Ratio You are given an array of strings arr[], where each arr[i] consists of lowercase english alphabets. You need to find the number of balanced strings in arr[] which can be formed by concatinating one or more contiguous strings of arr[]. A balanced string contains the equal number of vowels and consonants.	Take Home

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		Examples: Input: arr[] = ["aeio", "aa", "bc", "ot", "cdbc"] Output: 4 Explanation: arr[0..4], arr[1..2], arr[1..3], arr[3..3] are the balanced substrings with equal consonants and vowels. Input: arr[] = ["ab", "be"] Output: 3 Explanation: arr[0..0], arr[0..1], arr[1..1] are the balanced substrings with equal consonants and vowels. Input: arr[] = ["tz", "gfg", "ae"] Output: 0 Explanation: There is no such balanced substring present in arr[] with equal consonants and vowels. Link: https://www.geeksforgeeks.org/problems/balancing-consonants-and-vowels-ratio/1?page=1&category=Strings&sortBy=latest	